Title: Detecting Deepfakes Across Images, Video, and Audio

With rapid advances in generative AI, synthetic media can now convincingly mimic real people in images, videos, and voice recordings. Such deepfakes pose serious risks including misinformation, financial fraud, identity theft, reputational damage, and threats to public trust. Detecting manipulated or AI-generated media has therefore become a critical cybersecurity and digital trust challenge.

Problem Statement:

Design and develop an intelligent system capable of detecting deepfake or synthetically generated content across **one or more media types — images, video streams/recordings, and/or audio clips**. The solution should analyze the input media and determine whether it is authentic or manipulated, providing a confidence score and, if possible, indicators of why the content was flagged.

Participants may focus on a single modality (image, video, or audio) or build a multimodal solution that supports multiple types of media.

Expected Capabilities

The system should be able to:

- Detect AI-generated or manipulated **images**
- Identify deepfake manipulation in **video recordings or live streams**
- Detect synthetic or cloned **voice/audio samples**
- Output an authenticity assessment (e.g., Real / Suspected Deepfake)
- Provide a confidence level or explanation of findings (optional but encouraged)

Suggested Features (Optional)

Teams may choose to implement additional capabilities such as:

- Real-time detection for live media
- Visualization of suspicious regions or artifacts
- Support for multiple file formats
- User-friendly interface (web, mobile, or desktop)
- API-based detection service

- Reporting or audit trail of analyzed media

Evaluation Focus

Solutions will be assessed on:

- Detection accuracy and robustness
- Ability to handle real-world scenarios
- Innovation and technical approach
- Usability and scalability
- Performance within the hackathon timeframe